Homework 4

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Name: YuanHao Tang Student ID: 3120265621

Problem 1

Find out the Smith canonical form of the λ-matrix

$$A(\lambda) = \begin{pmatrix} 1-\lambda & \lambda^2 & \lambda \\ 2\lambda - 1 & \lambda(1-\lambda) & -2\lambda \\ 1+\lambda^2 & \lambda^2 & -\lambda^2 \end{pmatrix}.$$

Solution

Let D_k be the monic gcd of all $k \times k$ minors, with $D_0 = 1$. Then the invariant polynomials are $d_k = D_k/D_{k-1}$. Since $2(1-\lambda) + (2\lambda - 1) = 1$, $D_1 = 1$. Columns two and three are divisible by λ, so every second-order minor is divisible by λ. Moreover,

$$\begin{vmatrix} 1-\lambda & \lambda \\ 2\lambda - 1 & -2\lambda \end{vmatrix} = -\lambda,$$

giving $D_2 = \lambda$. Since $\det A = -\lambda^2(\lambda + 1)$, $D_3 = \lambda^2(\lambda + 1)$. Thus

$$\boxed{S_A(\lambda) = \operatorname{diag}\big(1, \lambda, \lambda(\lambda+1)\big)}.$$

Problem 2

Let $A(\lambda)$ be the $n \times n$ λ-matrix

$$A(\lambda) = \begin{pmatrix} \lambda - a & b_1 & 0 & \cdots & 0 \\ 0 & \lambda - a & b_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda - a & b_{n-1} \\ 0 & \cdots & \cdots & 0 & \lambda - a \end{pmatrix},$$

where $b_1, b_2, \ldots, b_{n-1}$ are all nonzero constants. Find out the invariant polynomials of $A(\lambda)$.

Solution

For $1 \le k < n$, the minor on rows $1, \ldots, k$ and columns $2, \ldots, k+1$ is the nonzero constant $b_1 \cdots b_k$, so $D_k = 1$. Also $D_n = \det A = (\lambda - a)^n$. Therefore

$$\boxed{d_1 = \cdots = d_{n-1} = 1, \qquad d_n = (\lambda - a)^n}.$$

Problem 3

Find out the Smith canonical form of the λ-matrix

$$A(\lambda) = \begin{pmatrix} \lambda & 0 & \cdots & 0 & a_0 \\ -1 & \lambda & \ddots & \vdots & a_1 \\ 0 & -1 & \ddots & 0 & \vdots \\ \vdots & \ddots & \ddots & \lambda & a_{n-2} \\ 0 & \cdots & 0 & -1 & \lambda + a_{n-1} \end{pmatrix}.$$

Solution

For $1 \le k < n$, the minor on rows $2, \dots, k+1$ and columns $1, \dots, k$ equals $(-1)^k$, hence $D_k = 1$. Expanding along the last column gives

$$D_n = \det A = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0 =: p(\lambda).$$

Consequently,

$$\boxed{S_A(\lambda) = \operatorname{diag}\big(\underbrace{1, \dots, 1}_{n-1}, p(\lambda)\big)}.$$

Problem 4

With respect to a matrix $A \in \mathbb{F}^{n\times n}$ and an indeterminate λ, $\lambda E - A$ is called the eigenmatrix, where E is the identity matrix. Find out the Smith canonical form of the eigenmatrix of the following matrix

$$A = \begin{pmatrix} -7 & -12 & -6 \\ 3 & 5 & 3 \\ 3 & 6 & 2 \end{pmatrix}.$$

Solution

If $\operatorname{char}\mathbb{F} \neq 3$, including the real and complex fields, put

$$B(\lambda) = \lambda I - A = \begin{pmatrix} \lambda + 7 & 12 & 6 \\ -3 & \lambda - 5 & -3 \\ -3 & -6 & \lambda - 2 \end{pmatrix}.$$

The nonzero constant entry -3 gives $D_1 = 1$. Since $B(-1)$ has rank one, every second-order minor is divisible by $\lambda + 1$, while

$$\begin{vmatrix} \lambda + 7 & 6 \\ -3 & -3 \end{vmatrix} = -3(\lambda + 1).$$

Thus $D_2 = \lambda + 1$. Also $D_3 = \det B = (\lambda + 1)^2(\lambda - 2)$, so

$$\boxed{S_B(\lambda) = \operatorname{diag}\big(1, \lambda + 1, (\lambda + 1)(\lambda - 2)\big)}.$$

If $\operatorname{char}\mathbb{F} = 3$, then $A = -I$ and the Smith form is $\boxed{(\lambda + 1)I_3}$.